ACADEMIC PORTFOLIO

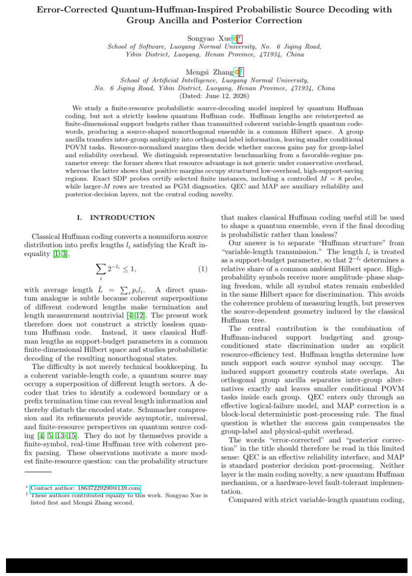
Songyao Xue

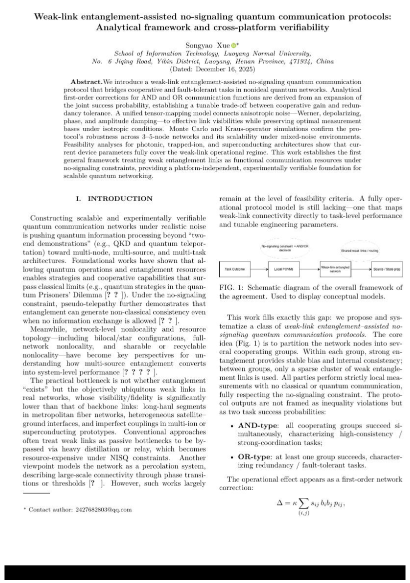
Software Engineering → Quantum Information

How can imperfect quantum resources be transformed into reliable computational advantages?

Quantum Information · Quantum Communication
Quantum Cryptography · Quantum Networks
Quantum Games · Quantum Machine Intelligence

Luoyang Normal University | Fall 2027 Applicant





Research Identity

A problem-driven trajectory from secure protocols to adaptive quantum networks

CENTRAL QUESTION

How can imperfect quantum resources be transformed into reliable computational advantages?

My research treats noise, limited entanglement, protocol exposure, and finite computational budgets not only as limitations, but as structures that can be modeled, allocated, and optimized.

RESEARCH PREPARATION

- B.Eng. in Software Engineering; academic rank in the top 10%.
- Supervisor-led reading group in quantum information, computing, cryptography, and communication.
- Python, Qiskit, PennyLane, numerical simulation, optimization, and LaTeX research workflows.
- Experience converting analytical models into reproducible code, figures, tables, and manuscripts.

RESEARCH TRAJECTORY

01	**Protocol security** How does system structure determine whether quantum primitives remain secure?
02	**Finite-resource decoding** When does a raw performance gain survive explicit resource accounting?
03	**Weak-link communication** Can noisy and heterogeneous entanglement links become functional resources?
04	**Adaptive quantum games** Can distributed quantum networks learn task-aware decision policies?
05	**Future Quantum Internet** How can communication, computation, security, and learning be optimized together?

03. Structural Security of Quantum Signatures

Quantum primitives are secure only when the full protocol architecture is secure.

RESEARCH PROBLEM

Grover operations, teleportation, and QKD may each be legitimate primitives, yet their composition can expose verification structure, keys, and message origin.

CORE APPROACH

Analyze complete information flow; abstract reusable inspection points; redesign the architecture with arbitration, strengthened QOTP, and teleportation.

MY CONTRIBUTION

Organized mathematical framework and notation; supported four reusable inspection points; implemented Qiskit verification; contributed to the strengthened construction and manuscript.

STATUS **Second author · Revised and resubmitted**

(AE1) In Step (AT4), Eve intercepts the first sequence of particles of the message qubits $U_{S_{AT}}|M\rangle$ and the message that Alice tells TC the locations of half the decoy qubits.

(AE2) In Step (AT5), Eve performs measurement on the decoy qubits using Z or X basis, and impersonates TC to tell Alice her measurement bases and measurement outcomes.

(AE3) In Step (AT6), Eve intercepts the other sequence of the particles that Alice sends to TC and the message that Alice tells TC the positions of decoy qubits and the types of unitary operations.

(AE4) In Step (AT8), Eve makes the unitary operator $U_V = 2|M\rangle\langle M| - I$ based on the known message. Then, she implements U_V on the received message to recover the shared key as $U_V U_{S_{AT}}|M\rangle = \pm|K_{AT}\rangle$. Thereby, Eve can obtain the shared key K_{AT} successfully.

In the Process between Eve and TC, Eve performs the following two steps to pass TC's verify by posing as Alice.

(ET1) Eve pretends to be Alice to announce forged message m' and makes its signature $U_{S_{AT}}|M'\rangle$ with K_{AT}.

(ET2) Eve impersonates Alice to communicate with TC by performing Steps (AT4)-(AT8).

Finally, there is nothing Eve need to do in the process between TC and Bob.

By performing the MITM attack above, $U_{S_{AT}}|M'\rangle$ as the signature of the forged message m' can pass TC's verifying. Therefore, Eve can forge Alice's signature for any message successfully.

Remark 1. Similarly, to forge Alice's signature in YKLY'2015 [4], Eve can perform an MITM attack between TC and Bob: Eve intercepts the message qubits $U_{S_{TB}}|M\rangle$ to obtain the shared key K_{TB}, pretends to be Alice to announce forged message m', makes the forged signature $U_{S_{TB}}|M'\rangle$ with K_{TB}, and sends $U_{S_{TB}}|M'\rangle$ to Bob as TC in YKLY'2015. In particular, if the attacker is Alice, her disavowal can be achieved.

SELECTED EVIDENCE

- MITM analysis identifies key recovery, forgery, and repudiation pathways.
- Four inspection points turn scheme-specific attacks into reusable protocol checks.
- The redesigned construction separates verification authority and strengthens signer-message binding.

NEXT QUESTION Security analysis showed that reliable advantage depends on system-level resource roles. This led to the question of how limited quantum resources should be allocated and evaluated.

04. Resource-Efficient Probabilistic Quantum Decoding

A performance gain is meaningful only when it survives the cost of ancilla, redundancy, and decoding.

RESEARCH PROBLEM

Can classical source statistics guide finite-dimensional quantum state design without claiming a strictly lossless coherent quantum Huffman code?

CORE APPROACH

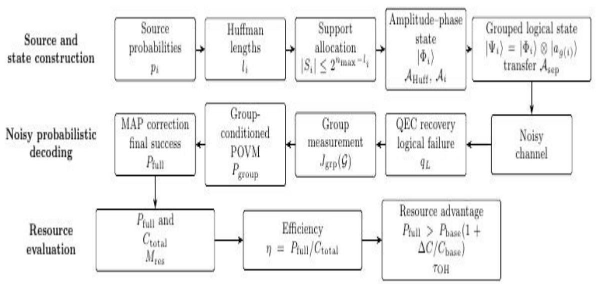
Reinterpret Huffman lengths as support budgets; construct source-shaped nonorthogonal ensembles; use group ancilla and conditional POVMs; evaluate QEC and MAP under explicit resource-normalized margins.

MY CONTRIBUTION

Developed the support-budget construction, group-conditioned discrimination, analytical conditions, SDP certificate logic, scalable simulations, and sensitivity analysis.

STATUS **First author · Manuscript in preparation**

$\lvert\Phi_i\rangle$	Payload state supported on S_i.
Γ_{ij}	Payload Gram-matrix entry, $\Gamma_{ij} = \langle\Phi_i\lvert\Phi_j\rangle$.
$g(i)$	Group index of symbol s_i.
$\mathcal{G}_g$ or G_g	The g-th group of source symbols.
G	Number of groups.
$\lvert a_g\rangle$	Orthogonal group-ancilla label.
$\lvert\Psi_i\rangle$	Grouped state $\lvert\Phi_i\rangle \otimes \lvert a_{g(i)}\rangle$.
$\mathcal{A}_{\rm Huff}$	Huffman-induced ambiguity budget.
$\mathcal{A}_i$	Symbol-level ambiguity contribution.
$\mathcal{A}_{\rm global}$	Total source-weighted overlap ambiguity before grouping.
$\mathcal{A}_{\rm in}$	Ambiguity remaining inside conditional group tasks.
$\mathcal{A}_{\rm sep}$	Ambiguity removed by the group label, $\mathcal{A}_{\rm global} - \mathcal{A}_{\rm in}$.
$P^{\star}_{\rm group}$	Optimal group-conditioned success probability.
$P_{\rm full}$	Effective full success after QEC reliability and fallback.
$P_{\rm base}$	Generic baseline success probability.
$P_{\rm base,opt}$	Numerical exact or representative baseline success.
q_L	Effective logical failure probability after recovery.
$P_{\rm fallback}$	Success probability of the fallback decision rule.
C_A	Group-label cost, $C_A = \lceil \log_2 G \rceil$.
$C_{\rm QEC}$	Modeled QEC redundancy cost.
$C_{\rm total}$	Total modeled resource cost.
η	Resource-normalized efficiency, $P_{\rm success}/C_{\rm total}$.
$M_{\rm res}$	Resource margin from Eq. (34).
$M_{\rm cert}$	SDP-certified normalized margin from Eq. (48).



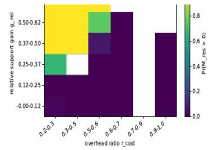


FIG. 6. Favorable-regime sweep, not the default benchmark. The heatmap reports the binned positive-margin fraction for no-QEC PGM-diagnostic rows as a function of relative support gain and overhead ratio. Positive regions appear where support gain is large enough and modeled overhead is low enough; this maps nonempty structured regimes, not universal advantage.

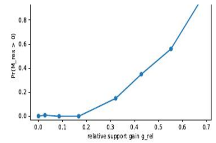


FIG. 7. Favorable-regime sweep probability versus relative support-budget gain $g_{\rm rel}$. Bars are binned PGM diagnostics with bin counts annotated in the generated figure and CSV. Positive margins become more likely at larger support savings, but this statistic is separate from the conservative default benchmark and is not an SDP certificate.

`favorable_regime_sweep.csv`. This sweep intentionally explores design degrees of freedom such as source skew, support-overlap shift, payload-generation mode, grouping strategy, group number, QEC overhead, and weighted overhead parameters. Its positive fraction is therefore not a typical-case success probability. It is a design-space map used to test whether positive-margin regions are nonempty and structurally interpretable.

The sweep contains 56160 PGM-labeled diagnostic rows generated by the current support-budget payload pipeline. The all-row positive fraction is $6828/56160 = 0.122$, the no-QEC subset has $4812/18720 = 0.257$ positive rows, and the weighted-cost margin has positive fraction 0.347. Positive no-QEC rows have larger mean support gain and lower overhead than non-positive rows: mean $G_{\rm comp} = 2.52$ versus 0.56, mean $g_{\rm rel} = 0.577$ versus 0.182, mean $r_{\rm sep} = 0.834$ versus 0.788, and mean $r_{\rm cost} = 0.356$ versus 0.565. Thus positive margins concentrate where support-budget saving, ambiguity transfer, and low overhead are present simultaneously.

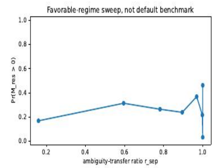


FIG. 8. Favorable-regime sweep probability versus ambiguity-transfer ratio $r_{\rm sep}$. The binned PGM diagnostic is not monotone enough to support a threshold law: ambiguity transfer helps, but positive margins also require support saving and low modeled overhead.

D. Exact certificate statistics

SELECTED EVIDENCE

- Default benchmarks show that reliability does not automatically imply resource advantage.
- Favorable-regime sweeps identify low-overhead, high-support-saving regions.
- Exact finite-instance SDP probes provide sufficient certificates rather than universal claims.

NEXT QUESTION The framework suggested that imperfect resources can be reshaped, not merely compensated, motivating weak entanglement as a task-aware network resource.

Framework pipeline and favorable-regime diagnostics from the manuscript.

05. Weak-Link Quantum Communication

Weak entanglement can become useful when its strength, orientation, and topology are aligned with a task.

RESEARCH PROBLEM

Real networks contain heterogeneous, noisy, and weak inter-group links. Can these links be used as functional resources rather than treated only as defects?

CORE APPROACH

Model cooperating groups as a weak-link network; derive first-order AND/OR corrections; map isotropic and anisotropic noise to effective visibilities; add Bayesian weighting and calibration rules.

MY CONTRIBUTION

Proposed the weak-link task model, sign and basis strategies, analytical corrections, noise mapping, parameter scans, cross-platform feasibility analysis, and simulation workflow.

STATUS **First author · Manuscript in preparation**

$$\mathbb{E}[s_{ij}] = 2\alpha - 1,$$
$$\mathbb{E}[\Delta(\mathbf{s})] = \kappa(2\alpha - 1)\sum_{(i,j)} b_i b_j p_{ij} \quad (12)$$

By tuning α, one can continuously shift the AND/OR trade-off to suit different task requirements (strong consistency vs. strong fault tolerance).

G. Decoding and Decision: A Bayesian Layer

Under strict no-signaling, decisions can only rely on locally permitted post-processing of each group's measurement outcomes. We adopt a lightweight Bayesian decoding layer: for a given task (AND/OR), a receiver (or an evaluator used for statistical partitioning) forms a likelihood ratio

$$\Lambda(\mathbf{a} \mid \mathbf{x}) = \frac{P(\mathbf{a} \mid \text{hypothesis } H_1, \mathbf{x})}{P(\mathbf{a} \mid \text{hypothesis } H_0, \mathbf{x})}$$

and makes a binary decision at a threshold τ, or performs maximum a posteriori (MAP) decisions in multi-message scenarios. Link heterogeneity (different p_{ij}) and alignment errors (κ) enter the likelihood via the statistics induced by Eqs. (8)–(10), thereby yielding decisions consistent with link quality. This decoding layer uses only allowable local statistics and pre-calibrated parameters, requiring no additional communication.

H. Perturbative Characterization of Visibility vs. Capacity/Error

To quantify the "weak-link penalty," introduce the small parameters $\varepsilon_{ij} := 1 - p_{ij}$. Under appropriate regularity conditions and a fixed measurement/decision architecture, the communication mutual information $I(m;\hat{m})$ (or an equivalent error exponent) can be written as

$$I(m;\hat{m}) \geq I_{\mathrm{EA}}^{(\mathrm{ideal})} - C_1\varepsilon - C_2\varepsilon^2 + o(\varepsilon^2) \quad (13)$$

comparison, the parameterization of this chapter will be used in later simulations and experiments under the following protocol:

1. **Visibility grid:** $p_{ij} \in \{p_{\min} : \Delta p : p_{\max}\}$, with a typical range $[0.75, 0.95]$;
2. **Alignment factor:** $\kappa \in \{0.80, 0.85, 0.90, 0.95, 1.00\}$;
3. **Basis strategies:** deterministic optimum $s_{ij}^\star$ (Eq. (11)) and randomized mixture $s_{ij} \sim \mathrm{Bernoulli}(\alpha)$ (Eq. (12));
4. **Statistical evaluation:** for each parameter point, repeat N independent trials to record $\widehat{P}_{\mathrm{AND}}$ and $\widehat{P}_{\mathrm{OR}}$, and provide confidence intervals (e.g., Clopper–Pearson) with multiple-comparison correction (FDR/Bonferroni) to validate the effective domain of Eqs. (9)–(10);
5. **Platform mapping:** photonic / trapped-ion / superconducting platforms map hardware-calibrated physical quantities (wave-plate angles, gate fidelities, crosstalk strengths) to (p_{ij}, κ) and generate task curves and phase diagrams under a unified script.

J. Summary

This chapter establishes a multi-group quantum communication framework under the *no-signaling* constraint, targeting **task-level indicators (AND/OR)**. The key ingredients are:

1. Unifying inter-group resources and noise via ρ_{net} and the correlation tensor T';
2. Capturing alignment errors through the geometric factor κ;
3. Linking *weak links—basis strategies—task performance* via the first-order correction Δ (Eq. (8));
4. Analytically characterizing the weak-link penalty

TABLE I: Noise–visibility–effective weak-link strength correspondence table (per-link evaluation)

Noise model	Axial visibilities (v_x, v_y, v_z)	Basis choice s_{ij}	Effective strength $p_{ij}^{\mathrm{eff}}(s_{ij})$	Substitution
Werner	$v_x = v_y = v_z = p$	arbitrary	p	(32), (35)
Depolarizing	$v_x = v_y = v_z = (1-q)^2$	arbitrary	$(1-q)^2$	(32), (35)
Phase damping	$v_x = v_y = (1-2d)^2,\ v_z = 1$	$+1 \to X,\ -1 \to Z$	$\frac{1+s_{ij}}{2}(1-2d)^2 + \frac{1-s_{ij}}{2}\cdot 1$	(32), (35)
Amplitude damping	$v_x = v_y = (1-\eta),\ v_z = (1-\eta)^2$	same as above	$\frac{1+s_{ij}}{2}(1-\eta) + \frac{1-s_{ij}}{2}(1-\eta)^2$	(32), (35)

[a] For each link, measure the visibilities $\hat{v}_x, \hat{v}_z$ and the alignment factor $\hat{\kappa}$. Select the optimal basis $s_{ij}^\star$ according to the task rule (Eqs. (37)/(38)), compute the effective strength p_{ij}^{eff}, substitute into Eqs. (33)–(34) to obtain the predicted value of $P_{\mathrm{AND/OR}}$. Uncertainty is given by Eq. (47).

TABLE II: Representative parameter ranges and performance thresholds.

Platform / scenario	κ	Target gain δ	S_b (experimental estimate)	Isotropic threshold $p \geq \frac{\delta}{\kappa S_b}$	Predicted gain $\kappa p S_b$
Photonic (SPDC)	0.90	0.05	0.60	$p \geq 0.0926$	0.324
Ion trap	0.95	0.05	0.50	$p \geq 0.1053$	0.238
Superconducting	0.93	0.05	0.55	$p \geq 0.0982$	0.282

[a] The parameter δ is the minimum desired AND/OR gain above the baseline (choose either task); S_b is inferred from experimental data without inter-group weak links; the column "Predicted gain" gives $\kappa p S_b$ for the specified (κ, p, S_b). In an actual experiment, these illustrative values should be replaced by the measured S_b and the platform-calibrated (κ, p).

2. Only two-qubit states with **partial entanglement (weak entanglement)** must be generated in a controllable way;
3. Measurement operations are standard single-qubit rotations and projective readout.

Hence the scheme is implementable on the three mainstream platforms: photonics, trapped ions, and superconducting circuits.

The physical parameters (p, κ) have the following meanings:

$$p = \text{visibility of the entangled link,}$$
$$\kappa = \text{measurement-basis alignment factor.} \quad (48)$$

• p is limited by source purity, decoherence, and detection noise;
• κ is governed by basis-setting errors, interferometer phase stability, or gate-calibration accuracy.

The experimental goal is to validate the analytical formulas of Secs. II–IV. The primary observables are

$$P_{\mathrm{AND}},\ P_{\mathrm{OR}},\ G_{\mathrm{AND/OR}} = \frac{P_{\mathrm{AND/OR}} - P_{\mathrm{AND/OR}}^{(0)}}{P_{\mathrm{AND/OR}}^{(0)}}, \quad (49)$$

B. Photonic Implementation

a. B.1 Entangled source and weak-link preparation. A β-BBO (or PPKTP) crystal pumped at the second-harmonic generates polarization-entangled pairs via SPDC, preparing $|\Psi^+\rangle = (|HV\rangle + |VH\rangle)/\sqrt{2}$. To obtain controllable weak entanglement:

1. Insert a polarization dephaser or liquid-crystal retarder in one arm to realize a Werner mixture

$$\rho_W(p) = p\,|\Psi^+\rangle\langle\Psi^+| + \frac{1-p}{4}\mathbb{I}_4,\ p \in [0.75, 0.95] \text{ tunable,} \quad (50)$$

2. Alternatively, use a Mach–Zehnder or a Sagnac loop to introduce phase mismatch and emulate phase-damping behavior.

Citation note. Cite standard SPDC entangled-source implementations and Sagnac/MZI entanglement sources, e.g., [? ? ?].

b. B.2 Measurement and basis alignment. Single-qubit bases are set by HWP/QWP stacks. A controlled angular offset $\delta\theta$ realizes an effective alignment factor $\kappa = \cos(2\delta\theta)$: e.g., $\delta\theta = 10^\circ$ gives $\kappa \simeq 0.94$.

SELECTED EVIDENCE

- The same variables support coordination-oriented AND and redundancy-oriented OR tasks.
- Four representative noise models are expressed through a common visibility framework.
- Photonic, trapped-ion, and superconducting mappings connect theory with tunable parameters.

NEXT QUESTION Representing weak links as weighted edges raised a system-level question: how should nodes adapt when topology, noise, cooperation, and competition change together?

Analytical task curves, robustness diagnostics, and platform mapping.

06. Learnable Weak-Link Multi-Group Quantum Games

Analytical structure can constrain learning while trainable variables adapt to topology and noise.

RESEARCH PROBLEM

Can a weak-link quantum network become an adaptive decision system rather than a passive communication medium?

CORE APPROACH

Represent groups as vertices and weak resources as signed weighted edges; derive a game tensor, sign rules, advantage criteria, and spectral bounds; relax discrete structure into differentiable variables.

MY CONTRIBUTION

Proposed the unified analytical-learning framework; derived first-order and spectral structure; implemented automated search, repeated trials, batch comparisons, visualization, backend evaluation, and cloud-hardware checks.

STATUS **First author · Submitted to Quantum Machine Intelligence · Under editorial review**

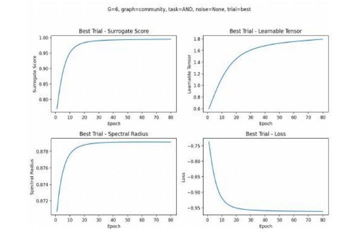


Fig. 1 Training dynamics of the selected best automated surrogate run under the six-group community-AND configuration. The curves report surrogate-level optimization quantities rather than theorem-level task values.

(a) Analytical network (b) Learned network

Fig. 2 Comparison between the analytical sign rule and the learned network structure for the selected six-group community-AND surrogate run.

7.3 Batch comparison across topology, task, and noise

The automated search result is complemented by a newly generated explicit batch protocol over graph type, task type, group count, and noise setting. Table 3 summarizes the highest-performing batch configurations after sorting by the final surrogate score.

The strongest explicit batch configurations reach $P_{\text{win}}^{\text{sur}} \approx 0.998$ under the current surrogate protocol. Because the updated batch is a single-seed implementation-side

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SELECTED EVIDENCE

- Selected surrogate runs remain aligned with analytical sign rules while adapting continuous parameters.
- Topology, task structure, and noise are evaluated through reproducible experiments.
- The work separates theorem-level results, surrogate objectives, and backend proxy quantities.

NEXT QUESTION This motivates quantum machine intelligence for future networks: learning should allocate links, measurements, routing, and redundancy while preserving interpretability.

Research Synthesis

One scientific question, four complementary answers

01 · SECURITY

Protocol structure determines whether a quantum primitive retains its security role.

Reliable advantage begins with authenticated information flow, protected verification authority, and composable security.

02 · RESOURCE ALLOCATION

Raw success must be normalized by logical and physical overhead.

Support geometry, conditional discrimination, and convex certificates identify when finite resources are genuinely useful.

03 · NETWORK FUNCTION

Weak links can be designed around a task rather than simply removed.

Visibility, basis alignment, topology, and noise become tunable variables in distributed quantum communication.

04 · ADAPTIVE DECISION

Learning can make network structure responsive without replacing analytical reasoning.

Differentiable models should operate inside physically meaningful and mathematically interpretable constraints.

I do not regard imperfect resources as obstacles to be removed before useful computation begins. I regard them as structures that can be modeled, allocated, certified, and exploited.

Methods and Technical Toolkit

Theoretical modeling, computational validation, and reproducible implementation

THEORY & MODELING

- Quantum protocol and information-flow analysis
- Graph-based weak-link network models
- POVM-based state discrimination
- Resource-normalized performance criteria
- First-order perturbative analysis
- Spectral and scalability bounds

OPTIMIZATION & COMPUTATION

- Python scientific computing and numerical experiments
- Qiskit and PennyLane circuit simulation
- Semidefinite programming and certificate logic
- Automated search and batch evaluation
- Noise, robustness, and sensitivity analysis
- Visualization and statistical summaries

RESEARCH ENGINEERING

- Reproducible experiment pipelines
- Artifact persistence and structured outputs
- LaTeX manuscript and appendix preparation
- Figures, tables, references, and notation consistency
- Git/Linux/Ubuntu development workflows
- Full-stack software and server deployment

SELECTED EVIDENCE OF PREPARATION

PREPARATION

Top 10% in Software Engineering; preparation in algorithms, data structures, linear algebra, probability, databases, and programming.

PREPARATION

Provincial mathematics and modeling awards; experience translating physical and engineering questions into computational models.

PREPARATION

Independent software development involving backend systems, web interfaces, Electron clients, and Ubuntu deployment.

Future Research Agenda

Toward a resource-aware, adaptive, and interpretable Quantum Internet

RESOURCE-AWARE NETWORK PROTOCOLS

Jointly optimize topology, routing, entanglement quality, security, and task performance under finite-resource constraints.

- Heterogeneous links
- Adaptive entanglement allocation
- End-to-end task metrics
- Resource-aware security

LEARNABLE QUANTUM NETWORKS

Develop graph-structured learning systems that adapt measurements and policies under calibrated noise while preserving analytical guarantees.

- Graph learning
- Variational optimization
- Bayesian decisions
- Convex certification

UNIFIED DISTRIBUTED ARCHITECTURES

Study communication, computation, cryptography, and error correction as interacting components consuming the same limited resources.

- Distributed quantum computing
- Quantum cryptography
- Error-aware orchestration
- Interpretable control

To become an independent researcher developing intelligent and resource-efficient quantum information systems that integrate communication, computation, security, and learning.

Selected Outputs and Contact

Current research record and application profile

MANUSCRIPTS AND PAPERS

On Grover-Based Quantum Signature Schemes
Second author · Revised and resubmitted

A Learnable Unified Framework for Weak-Link Multi-Group Quantum Games
First author · Under editorial review

Weak-Link Entanglement-Assisted No-Signaling Quantum Communication Protocols
First author · Manuscript in preparation

Error-Corrected Quantum-Huffman-Inspired Probabilistic Source Decoding
First author · Manuscript in preparation

HONORS

- Two Provincial Third Prizes, Chinese Mathematics Competition (2024, 2025).
- Provincial Third Prize, Contemporary Undergraduate Mathematical Contest in Modeling (2025).
- Honorable Mention, Shuwei Cup Mathematical Modeling Challenge (2026).

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Fall 2027

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RESEARCH FOCUS
Quantum Information · Communication · Cryptography · Networks

Quantum Machine Intelligence